

› Lab - 1 | 22 January 2026

↳ 6 cells hidden

› Lab - 2 | 29 January 2026

↳ 14 cells hidden

› LAB -3 5 Feb

↳ 2 cells hidden

› Lab- 4 | 12 Feb

↳ 20 cells hidden

✓ Lab - 5 | 19 Feb

```
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

y = np.array([0, 1, 1, 0])

model = Sequential()

model.add(Dense(4, input_dim=2, activation='relu'))

model.add(Dense(1, activation='sigmoid'))

model.compile(loss='binary_crossentropy',
```

```
optimizer='adam',  
metrics=['accuracy'])  
  
model.fit(X, y, epochs=500, verbose=0)  
  
predictions = model.predict(X)  
  
print("\nPredictions:")  
for i in range(len(X)):  
    print(f"Input: {X[i]} -> Predicted: {round(predictions[i][0])}")
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: Us  
super().__init__(activity_regularizer=activity_regularizer, **kwargs)  
1/1  0s 90ms/step
```

```
Predictions:  
Input: [0 0] -> Predicted: 0  
Input: [0 1] -> Predicted: 1  
Input: [1 0] -> Predicted: 1  
Input: [1 1] -> Predicted: 0
```

Lab 6 || 26 Feb

↳ 2 cells hidden

